

Manuel Weber

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Experienced Applied Scientist and Researcher with a background in particle physics and geospatial analytics, over a decade of expertise in statistical data analysis, and 8+ years in deep learning research specializing in computer vision and transformer-based models. Skilled in tackling complex challenges in fundamental physics, remote sensing, climate, and sustainability using petabyte-scale datasets. Proficient in distributed computing and large-scale model training. Led projects from inception to production and published in high-impact journals like *Nature* and *Physical Review Letters*.

Work experience

Research Scientist (Geospatial AI)

2024 - present

EarthDaily Analytics

- Conducting research on foundation models for earth observation; Developed an attention-based multi-sensor fusion mechanism to process mixed-resolution imagery (ICLR 2025, doi.org/10.48550/arXiv.2504.18770).
- Working on wildfire risk modelling with the use of GeoFMs.
- Developed methodology for assessing supply risk for various crops based on weather/climate data and spectral indices.

Applied Scientist (Machine Learning)

2019 - 2024

Descartes Labs

- Developed deep learning model for global prediction of aboveground biomass density, tree height and cover using multi-spectral, multi-modal satellite imagery and LiDAR data (Remote Sensing journal, doi.org/10.3390/rs17091594).
- Implemented transformer-based model for processing weather data time-series to predict perennial crop production.
- Initiated and led the development of a novel GAN-based approach for generic change detection in satellite imagery.
- Conducted research on neural architecture search and representation learning for satellite imagery.
- Implemented various computer vision models for the detection and tracking of commodity-carrying objects in video feeds.
- Built a solution for monitoring shipping activities at 1000+ ports worldwide using synthetic aperture radar.
- Lead cross-functional deep learning team.

Postdoctoral Fellow

2013 - 2018

Stanford University

- Researched novel deep learning and computer vision approaches for the energy and vertex reconstruction of particle interactions in LXe detectors.
- Led a team of 4 researchers to optimize regression algorithms for the event reconstruction and automate detector calibration pipeline.
- Supervised analysis teams and conducted rare event searches incorporating machine learning techniques.
- Initiated an R&D project for developing a new type of single-photon detector under a DARPA research grant. Demonstrated the use of organic liquids for the photon conversion.

Education

Ph.D. - Neutrino Physics

University of Bern

M.S. - Particle Physics

University of Bern

B.S. - Theoretical Physics

University of Bern / Lund

Skills

- Deep learning / ML
- Computer vision
- Statistical data analysis
- Time series analysis
- Distributed computing
- Cloud infrastructure (GCP/AWS)
- Geospatial data processing

Tools

Python, Tensorflow, Numpy, OpenCV, scikit-learn, SciPy, GDAL, Kubernetes, Vertex-AI

Data visualization and interaction with matplotlib, ipyleaflet, ipywidgets, bokeh

Publications and Patents

3 patents related to computer vision and geospatial data

21 data analysis, ML

publications in Nature, Phys Rev Lett, Phys Rev A, C, D, Remote Sens with 1000+ citations

12 technical publications

in JINST, JCAP, Rev Sci Instrum, Nucl Instrum Methods Phys with 100+ citations